

Theoretical complexity bounds for Bubble Sort

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Abstract

In this work, we consider Bubble sort. We derive a new tighter bounds on complexity reducing previously published bound on $\mathcal{O}(n^2)$ to $\mathcal{O}(n^2 - 1)$.

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1 Introduction

Bubble sort is very important algorithm, it is used in various applications. Theoretical bounds are useful for ...

2 Related work

Astrachan (2003) proved $\mathcal{O}(n^2)$ complexity bound. The $\mathcal{O}(n^2)$ bound was previously proven (Astrachan, 2003).

3 Complexity bounds

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Appendix A.

In this appendix we prove the following theorem from Section 6.2:

Theorem *Let u, v, w be discrete variables such that v, w do not co-occur with u (i.e., $u \neq 0 \Rightarrow v = w = 0$ in a given dataset $\mathcal{D}$). Let N_{v0}, N_{w0} be the number of data points for which $v = 0, w = 0$ respectively, and let I_{uv}, I_{uw} be the respective empirical mutual information values based on the sample $\mathcal{D}$. Then*

$$N_{v0} > N_{w0} \Rightarrow I_{uv} \leq I_{uw}$$

with equality only if u is identically 0. ■

Appendix B.

Proof. We use the notation:

$$P_v(i) = \frac{N_v^i}{N}, \quad i \neq 0; \quad P_{v0} \equiv P_v(0) = 1 - \sum_{i \neq 0} P_v(i).$$

These values represent the (empirical) probabilities of v taking value $i \neq 0$ and 0 respectively. Entropies will be denoted by H . We aim to show that $\frac{\partial I_{uv}}{\partial P_{v0}} < 0 \dots$

Remainder omitted in this sample. See <http://www.jmlr.org/papers/> for full paper.

References

Owen Astrachan. Bubble sort: an archaeological algorithmic analysis. *ACM Sigcse Bulletin*, 35(1):1–5, 2003.